Session 14 (AIML) – Assignment Question

Name of Student : Ziya Shaikh

Domain Name : AIML Development

College Name : Zeal Polytechnic, pune

Branch Name : AIML

DATE: 01.07.2026, WEDNESDAY

Q1:

Task 13.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py (3.13.5)

File Edit Format Run Options Window Help

```python
# Q1. Array Properties

import numpy as np

# Create 5x6 array with random integers
arr = np.random.randint(10, 101, (5, 6))

print("Array:")
print(arr)

# Properties
print("\nShape:", arr.shape)
print("Total Elements:", arr.size)
print("Data Type:", arr.dtype)

# End
```

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py ==
Array:
[[77 97 11 42 62 98]
 [62 89 18 51 71 46]
 [85 88 56 50 54 53]
 [86 36 91 29 74 69]
 [24 72 82 80 45 63]]

Shape: (5, 6)
Total Elements: 30
Data Type: int32
>>>
```

Q2:

Task 13.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py (3.13.5)

File Edit Format Run Options Window Help

```python
# Q2. Statistical Methods

import numpy as np

# Generate 20 random numbers
arr = np.random.randint(1, 51, 20)

print("Array:")
print(arr)

# Minimum and index
print("\nMinimum:", arr.min())
print("Index:", arr.argmin())

# Maximum and index
print("\nMaximum:", arr.max())
print("Index:", arr.argmax())

# Statistics
print("\nSum:", arr.sum())
print("Mean:", arr.mean())
print("Standard Deviation:", arr.std())

# End
```

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py ==
Array:
[ 2 15 20 50 41 21 32 20 45  4  7 39 30 29  8 49 43 35 11 15]

Minimum: 2
Index: 0

Maximum: 50
Index: 3

Sum: 516
Mean: 25.8
Standard Deviation: 15.164432069813891
>>>
```

Q3:

Task 13.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py (3.13.5)

File Edit Format Run Options Window Help

```python
# Q3. Statistical Methods on 2D Array

import numpy as np

# Create matrix
arr = np.random.randint(20, 81, (4, 5))

print("Matrix:")
print(arr)

print("\nMinimum:", arr.min())
print("Maximum:", arr.max())

print("Sum:", arr.sum())

print("Mean:", arr.mean())

print("Standard Deviation:", arr.std())

# Row Sum
print("\nRow Sum:")
print(arr.sum(axis=1))

# Column Sum
print("\nColumn Sum:")
print(arr.sum(axis=0))

# End
```

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py ==
Matrix:
[[38 23 52 20 61]
 [73 66 66 45 48]
 [62 42 71 67 56]
 [39 69 33 39 71]]

Minimum: 20
Maximum: 73
Sum: 1041
Mean: 52.05
Standard Deviation: 16.110477956907424

Row Sum:
[194 298 298 251]

Column Sum:
[212 200 222 171 236]
>>>
```

Q4:

Task 13.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/

File Edit Format Run Options Window Help

```python
# Q4. Reshape

import numpy as np

# Create array
arr = np.arange(1, 25)

print("Original:")
print(arr)

# Reshape
a = arr.reshape(4, 6)

b = arr.reshape(3, 8)

c = arr.reshape(2, 3, 4)

print("\n4x6")
print(a)

print(a.shape)

print("\n3x8")
print(b)

print(b.shape)

print("\n2x3x4")
print(c)

print(c.shape)

# End
```

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> 
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py ==
Original:
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24]

4x6
[[ 1  2  3  4  5  6]
 [ 7  8  9 10 11 12]
 [13 14 15 16 17 18]
 [19 20 21 22 23 24]]
(4, 6)

3x8
[[ 1  2  3  4  5  6  7  8]
 [ 9 10 11 12 13 14 15 16]
 [17 18 19 20 21 22 23 24]]
(3, 8)

2x3x4
[[[ 1  2  3  4]
  [ 5  6  7  8]
  [ 9 10 11 12]]

 [[13 14 15 16]
  [17 18 19 20]
  [21 22 23 24]]]
(2, 3, 4)
>>> 
```

Q5:

```
IDLE Shell 3.13.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py ==
Original Array:
[[ 10  20  30  40]
 [ 50  60  70  80]
 [ 90 100 110 120]]

First Row:
[10 20 30 40]

Last Column:
[ 40  80 120]

Center 2x2:
[[20 30]
 [60 70]]

Even Numbers:
[ 10  20  30  40  50  60  70  80  90 100 110 120]
>>>
```

Task 13.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py (3.13.5)

File Edit Format Run Options Window Help

```python
# Q5. NumPy Indexing

import numpy as np

# Create array
arr = np.array([
   [10, 20, 30, 40],
   [50, 60, 70, 80],
   [90, 100, 110, 120]
])

print("Original Array:")
print(arr)

# First row
print("\nFirst Row:")
print(arr[0])

# Last column
print("\nLast Column:")
print(arr[:, -1])

# Center 2x2 matrix
print("\nCenter 2x2:")
print(arr[0:2, 1:3])

# Even numbers
print("\nEven Numbers:")
print(arr[arr % 2 == 0])

# End
```

Q6:

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on v
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py ==
Original Matrix:
[[ 33  19  63 100  96]
 [ 42   2  44  85  44]
 [  7  33  88  47  36]
 [ 29  98  36  50  25]
 [ 33  97  56  30  64]]

Diagonal Elements:
[33  2 88 50 64]

Elements Greater Than 50:
[ 63 100  96  85  88  98  97  56  64]

Modified Matrix:
[[ 33   0  63 100  96]
 [ 42   0  44  85  44]
 [  0  33  88  47  36]
 [  0  98  36  50   0]
 [ 33  97  56  30  64]]
>>>
```

Task 13.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py (3.13.5)

File Edit Format Run Options Window Help

```python
# Q6. Advanced Indexing

import numpy as np

# Create matrix
arr = np.random.randint(1, 101, (5, 5))

print("Original Matrix:")
print(arr)

# Diagonal
print("\nDiagonal Elements:")
print(np.diag(arr))

# Greater than 50
print("\nElements Greater Than 50:")
print(arr[arr > 50])

# Replace less than 30
arr[arr < 30] = 0

print("\nModified Matrix:")
print(arr)

# End
```

Q7:

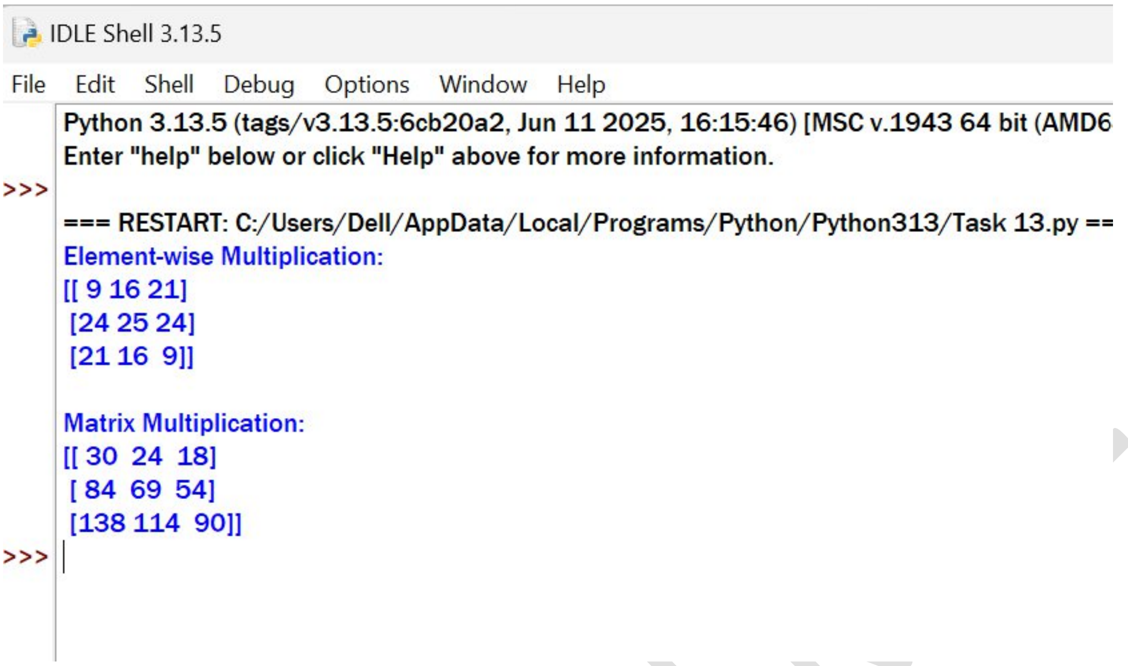
```
IDLE Shell 3.13.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py ==
Addition:
[11 22 33 44]

Subtraction:
[ 9 18 27 36]

Multiplication:
[ 10  40  90 160]

Division:
[10. 10. 10. 10.]

Power:
[      10     400   27000 2560000]

Modulo:
[0 0 0 0]
>>>
```

Task 13.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13

File Edit Format Run Options Window Help

```python
# Q7. Arithmetic Operations

import numpy as np

a = np.array([10, 20, 30, 40])
b = np.array([1, 2, 3, 4])

print("Addition:")
print(a + b)

print("\nSubtraction:")
print(a - b)

print("\nMultiplication:")
print(a * b)

print("\nDivision:")
print(a / b)

print("\nPower:")
print(a ** b)

print("\nModulo:")
print(a % b)

# End
```

Q8:


IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD6
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py ==
Element-wise Multiplication:
[[9 16 21]
[24 25 24]
[21 16 9]]
Matrix Multiplication:
[[30 24 18]
[84 69 54]
[138 114 90]]
>>>

Task 13.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Ta:

File Edit Format Run Options Window Help

```
# Q8. Matrix Multiplication

import numpy as np

A = np.array([
[1,2,3],
[4,5,6],
[7,8,9]
])

B = np.array([
[9,8,7],
[6,5,4],
[3,2,1]
])

# Element-wise multiplication
print("Element-wise Multiplication:")
print(A * B)

# Matrix multiplication
print("\nMatrix Multiplication:")
print(A @ B)

# Difference:
# * multiplies position-wise
# @ performs actual matrix multiplication

# End
```

Q9:

```
IDLE Shell 3.13.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py ==
Matrix:
[[ 1.00176914  0.1191138  -0.57787005  1.63824765 -0.22636542 -0.98002914]
 [ 0.35235541  0.83732011 -0.50485983  1.48724825 -0.46933102 -1.08076946]
 [-0.70801691  0.61480676  1.37699219  0.92650015  0.40287306 -0.67175359]
 [ 0.59694872 -1.10516709 -0.08503768  1.00430729 -0.61365499  0.79073033]
 [-2.74919943 -0.49980732 -0.44970344  1.8993786   1.15024951  0.14150674]
 [-1.84541942  1.69521464 -1.49162088  0.1959101  -1.95497223  0.32251749]]

Shape: (6, 6)
Size: 36
Data Type: float64

Max Index: 27
Min Index: 24

Top Left Matrix:
[[ 1.00176914  0.1191138  -0.57787005]
 [ 0.35235541  0.83732011 -0.50485983]
 [-0.70801691  0.61480676  1.37699219]]

Modified Matrix:
[[1.00176914 0.1191138  0.57787005 1.63824765 0.22636542 0.98002914]
 [0.35235541 0.83732011 0.50485983 1.48724825 0.46933102 1.08076946]
 [0.70801691 0.61480676 1.37699219 0.92650015 0.40287306 0.67175359]
 [0.59694872 1.10516709 0.08503768 1.00430729 0.61365499 0.79073033]
 [2.74919943 0.49980732 0.44970344 1.8993786  1.15024951 0.14150674]
 [1.84541942 1.69521464 1.49162088 0.1959101  1.95497223 0.32251749]]

Mean:
0.9046546620840452
>>>
```

Task 13.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Ta

File Edit Format Run Options Window Help

```python
# Q9. Combined Operations

import numpy as np

# Generate matrix
arr = np.random.randn(6, 6)

print("Matrix:")
print(arr)

# Properties
print("\nShape:", arr.shape)
print("Size:", arr.size)
print("Data Type:", arr.dtype)

# Indexes
print("\nMax Index:", arr.argmax())
print("Min Index:", arr.argmin())

# Top left 3x3
print("\nTop Left Matrix:")
print(arr[0:3, 0:3])

# Convert negatives to positive
arr = np.abs(arr)

print("\nModified Matrix:")
print(arr)

# Mean
print("\nMean:")
print(arr.mean())

# End
```

Q10:

```
IDLE Shell 3.13.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on wi
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py ==
Student Marks:
[[74 31 59 45 50]
 [58 84 50 99 43]
 [75 92 94 66 66]
 [63 40 78 41 93]
 [91 90 80 60 47]
 [36 50 85 89 40]
 [83 94 74 52 37]
 [50 83 82 63 88]
 [99 39 58 37 51]
 [90 47 35 85 98]]

Total Marks:
[259 334 393 315 368 300 340 366 284 355]

Average Marks:
[51.8 66.8 78.6 63.  73.6 60.  68.  73.2 56.8 71. ]

Top Student Index: 2
Lowest Student Index: 0

Class Mean: 66.28
Class Standard Deviation: 21.086526503907656

Marks of Top 3 Students:
[[50 83 82 63 88]
 [91 90 80 60 47]
 [75 92 94 66 66]]
>>>
```

Task 13.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 13.py (3.13.5)

File Edit Format Run Options Window Help

```python
# Q10. Student Performance Analysis

import numpy as np

# Generate marks
marks = np.random.randint(30, 101, (10, 5))

print("Student Marks:")
print(marks)

# Total marks
total = marks.sum(axis=1)

# Average
average = marks.mean(axis=1)

print("\nTotal Marks:")
print(total)

print("\nAverage Marks:")
print(average)

# Highest and lowest
high = total.argmax()
low = total.argmin()

print("\nTop Student Index:", high)

print("Lowest Student Index:", low)

# Class statistics
print("\nClass Mean:", marks.mean())

print("Class Standard Deviation:", marks.std())
```


Top 3 students


```python
# Top 3 students
top3 = np.argsort(total)[-3:]

print("\nMarks of Top 3 Students:")
print(marks[top3])

# End
```